Jay Dhumal

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SUMMARY

AI Engineer building production agentic AI and GenAI systems: multi-agent workflows, RAG chatbots, Text-to-SQL agents, enterprise LLM gateways and document-intelligence pipelines on Amazon Bedrock and Azure AI Foundry. Hands-on with LangGraph, LangChain, MCP and guardrailed LLM deployments across Claude, GPT and open-weight models.

TECHNICAL SKILLS

Agentic AI: LangGraph, LangChain, multi-agent systems, tool-calling agents, MCP, structured outputs (Pydantic), prompting
RAG & Retrieval: RAG pipelines, ChromaDB, OpenSearch vector store, embeddings, Text-to-SQL, schema-aware agents
LLM Platforms: Amazon Bedrock (Agents, Guardrails), Azure AI Foundry, Azure OpenAI, Claude, Groq, OpenAI APIs
LLM Governance: AI Gateway design, Entra ID policy validation, token-limit monitoring, PII and prompt-injection guardrails
Models: Claude (incl. Haiku 4.5), Amazon Titan, GPT family, open-weight models served via Groq
Multimodal & Document AI: Azure Document Intelligence, Textract, Rekognition, Transcribe, Comprehend
Build & Ship: Python, Chainlit, Streamlit, Flask, PostgreSQL (agent memory), Git

EXPERIENCE

Pentagon System and Services Dec 2024 – Present
AI Engineer, AI Services (Cloud Department) *Mumbai, Maharashtra*

- Designed an enterprise **AI Gateway** on Azure as the single governed entry point for all AI and LLM APIs, with custom policies, Entra ID validation and a per-user token-limit analytics dashboard; extended the same pattern to GitHub Copilot.
- **Integrated Claude Desktop and Claude CLI** traffic through the AI Gateway for a 15–20 user pilot with Azure Workbook monitoring; built on Azure API Management governing 400+ operations across 30+ APIs, including LLM requests.
- Deployed Azure Foundry models and securely connected them to **GitHub Copilot and Claude Code** in VS Code over a private tunnel for an enterprise client.
- Built a **Text-to-SQL GenAI agent** with Amazon Bedrock and LangChain that interprets natural-language questions, generates SQL and enables self-service analytics; **reduced query time by 50%**.
- Built a production **RAG chatbot over 800+ Statements of Work**, combining document retrieval with LLM-generated, context-aware responses for faster knowledge discovery.
- Contributed to a **supervisor multi-agent FinOps system** on Bedrock (cost analysis, optimization and resource-intelligence agents; RAG over OpenSearch Serverless; Guardrails; prompt caching to lower inference cost).
- Contributed to **Bedrock document agents on Claude Haiku 4.5** (RAG chatbot, document analytics, support bot) with Textract OCR, human-in-the-loop review and Guardrails for PII and prompt injection inside a private VPC.
- Contributed to **GenAI-driven underwriting** for an NBFC: Bedrock LLMs generate banking and bureau summaries and adaptive, borrower-specific underwriting questions, with a prompt-refinement feedback loop.
- Delivered **agentic AI for HR operations** (onboarding to exit interviews), a **RAG chatbot over CloudTrail logs** to monitor Amazon Q licence usage, and **agentic analytics on ticket-resolution data** for an operations team.
- Prepared case studies, architecture diagrams and evidence for the **AWS Gen AI & Agentic AI Competency audit (passed)**; worked with cross-functional teams on cloud AI solutions.

Nettech India May 2024 – Oct 2024
Data Scientist & AI/ML Trainer *Mumbai, Maharashtra*

- Trained **50+ participants** in Python, NLP and ML/AI (TF-IDF, BoW, projects) with **95%+ completion**.
- Delivered hands-on NLP and machine-learning instruction and guided learners through applied AI project development.

PROJECTS

Tangarin – Agentic Text-to-SQL System | Python, LangGraph, LangChain, Groq, PostgreSQL, Chainlit Feb 2026

- LLM agent that reasons over database context with schema-inspection and query-execution tools; PostgreSQL-backed conversation memory and authenticated Chainlit UI for self-service analytics.

Lore – Agentic Skills Assistant | LangGraph, Groq, ChromaDB, sentence-transformers, Streamlit In development

- Agent that decides when to invoke pluggable SKILL.md skills (two-stage intent classifier); routes uploaded documents between full-context and RAG by token count; sandboxed skill scripts produce real .docx output.

Covenant Extraction AI Agent | Azure Document Intelligence, Azure AI Foundry, Pydantic Sept 2026

- Extracted text from scanned Debenture Trust Deeds, passed it to a Foundry model for covenant discovery (for RBI/SEBI compliance checks) and enforced typed Pydantic schemas for LLM-ready outputs.

Multimodal Sentiment Analysis Pipeline | Amazon Comprehend, Rekognition, Transcribe Call Analytics Sept 2025

- Combined text, facial-expression and speech signals for a lending organization, synchronizing three AI services at sentence level into automated unified sentiment reports.

Inventra – MCP Tool Server + LangGraph Agent | MCP, LangGraph, Flask, PostgreSQL

- Multi-service Model Context Protocol system exposing JWT-secured tools over Streamable HTTP to a LangGraph agent.

Context Vault – AI Chat Memory Extension | Chrome extension, JavaScript

- Chrome extension that saves AI chat transcripts and reinjects them as context across ChatGPT and Claude sessions.

EDUCATION & CERTIFICATIONS

Mumbai University – Bachelor of Science in Computer Science Sept 2021 – May 2024
AI-103: Microsoft Certified: Azure AI Apps and Agents Developer Associate Sep 2026